Application of Explainable Artificial Intelligence (XAI) in Random Forest Forecasting for Analysis of Structural Changes in Jakarta Air Quality

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Abstract— Jakarta, as a metropolitan city with intensive economic activities and high mobility, faces significant air pollution challenges. This condition highlights the need for analytical methods that can not only forecast air quality but also explain the factors influencing prediction outcomes. This study utilizes Jakarta Air Pollution Standard Index (ISPU) data from 2016–2025, covering five monitoring stations with a total of 12,247 daily observations. The analysis employs an Independent Sample T-Test to identify structural changes in air quality following the COVID-19 pandemic, Random Forest for PM10 concentration forecasting, and SHAP (SHapley Additive exPlanations) as an Explainable Artificial Intelligence (XAI) approach to interpret the model. The results indicate a significant structural change after 2020 with a p-value of 0.00026. The Random Forest model achieved an RMSE of 12.19, MAE of 9.34, and R^2 of 0.41, while demonstrating stable performance under Time-Series Cross-Validation. SHAP analysis revealed that PM10(t-1) was the most influential feature affecting prediction outcomes. Overall, the integration of Random Forest and XAI produced an air quality forecasting model that is reasonably accurate, stable, and transparent, making it a promising tool for supporting data-driven decision-making in urban air pollution control.

Keywords— air quality, PM10, Random Forest, Explainable Artificial Intelligence, SHAP, forecasting.

I. INTRODUCTION

Being a city with a high population density, Jakarta's traffic activities will certainly be one of the main contributors to the highest air pollution in the world [1]. Based on observations from the Jakarta Air Pollution Standard Index, PM10 was recorded as one of the most dominant polluting particulates during the observation period 2016-2025 [2]. These observations are carried out every day, recorded and published so that they can be easily accessed by the public. The results of data observations carried out consistently every day will certainly produce a lot of data. This abundant data opens up opportunities for utilizing data analytics to support decision making. From this idea, the urgency arises to carry out data management, an analytical system that is not only able to predict but can also explain the logic of its predictions to policy makers. This high urgency is also related to the population in Jakarta, which is occupied by approximately 10.6 million people [3]. Health data in DKI Jakarta Province in 2022 shows that the 10 largest diseases, such as hypertension, ARI and pneumonia (Bappenas 2023), are related to the impact of air pollution [2].

The changes that occurred in 2020 due to COVID-19 succeeded in becoming a structural change for the model machine learning whichever is accustomed to and trained using pre-pandemic patterns. The implementation of Large-Scale Social Restrictions (PSBB) in March 2020 has proven to significantly reduce the mobility and volume of land vehicles, which has resulted in a drastic reduction in pollutant levels in Jakarta [4] These emissions reductions set records that were difficult for any previous traffic regulation intervention to achieve, pollutant gases such as carbon monoxide (CO) and particulates dropped significantly due to reduced mobilization, rather than due to technological updates [4]. When viewed from the perspective of environmental research methodology, this phenomenon is often referred to as natural experiment, that is, the condition when an unplanned (external) event occurs accidentally creates exogenous causal variations in the variables studied [5]

Decision to use architecture Random Forest specifically to reduce and prevent the presence of distribution anomalies through merging procedures (ensamble) of several decision trees. Naturally, architecture Random Forest tends to be less able to recognize temporal autocorrelation, to reduce this limitation a lag variable is used which is definitively a time shift between a variable and itself, the variable used is (t-1, t-2, t-7) which explains data on short-term temporal dependence and Jakarta's weekly pattern [6]. Autocorrelation of model robustness (robustness model) is also the basis for comparison and consideration with other models [7].

This research is here to fill and connect gaps in research and studies first which tend to make predictions only using conventional statistical models, or not integrate the XAI interpretation layer [8]. The research combines three mutually correlated methodologies: Random Forest, T-Test, and SHAP.

The hope is that the integration of these three methods can build comprehensive results by combining predictive and prescriptive aspects, in addition to providing an empirical basis for development Decision Support System (DSS) transparent, validable and accountable air quality.

II. RESEARCH METHODS

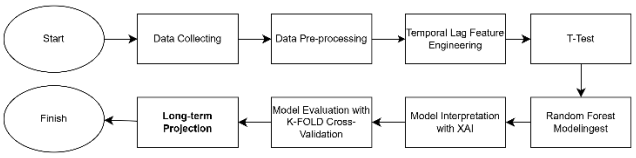


Figure 1 Research Method Flowchart

A. Data Collection into Datasets

The analysis focused on the five stations in Jakarta, in order to get a geographical picture of Jakarta's air quality. The data obtained comes from a combination of annual Air Pollution Standard Index data (from 2016-2025) published by Satu Data Indonesia, which can be accessed via the official website https://data.go.id. The data contained all five observation stations, with daily recordings yielding a total of 12,247 daily data. An explanation of the dataset is explained more deeply through **Table 1.**

TABLE 1 Dataset and Description

VARIABLE NAME	DESCRIPTION
Data Period	Time period for recording and reporting monthly data
Date	Daily recording time
Station	Monitoring location
PM10	Particulate Matter 10 measures the concentration of dust, dirt, or aerosol particles in air having a diameter of 10 micrometers or smaller [9]
PM2.5	Particulate Matter 2.5 measures the concentration of very fine air particles with a diameter of 2.5 micrometers or smaller [9]
SO2	Sulfur Dioxide gas which is generally produced from the combustion activity of fossil fuels containing sulfur [9]
CO	Carbon Monoxide is a colorless and odorless gas that comes from incomplete combustion [9]
O3	Air Surface Ozone is a secondary pollutant that is not emitted directly, but is formed through complex photochemical reactions between sunlight, [9]
NO2	Nitrogen Dioxide is a toxic and highly reactive gas that arises predominantly from high temperature combustion processes [9]
Max	Maximum Air Pollutant Standard Index (ISPU) value [9]
Critical	Critical Pollutant Parameters that show which pollutants contribute the highest value[9]

With a retarded observation time from January 1, 2016, to November 30, 2025. The data collected becomes this dataset is a recording taken from five monitoring areas, namely DKI1 – Central Jakarta (Bundaran HI), DKI2 – North Jakarta (Kelapa Gading), DKI3 – South Jakarta (Jagakarsa), DKI4 – East Jakarta (Lubang Buaya), DKI5 – West Jakarta (Kebon Jeruk). Although the dataset contains some pollutants, the study focused on PM10 as a target variable and thus other variables were not used in the modelling process.

B. Data Pre-Processing

The data pre-processing process is designed to turn raw data that tends to be messy and difficult to understand, into a data set that is more organized and ready to be processed [10]. Data cleaning starts by handling the format, then cleaning the anomalies found. The pre-processing stage begins with data from an excel file entitled "DATA ISPU JAKARTA.xls" which is extracted into a data frames named jkt. Frame data this facilitates further data processing and analysis. Then, the date column is then extracted into three separate columns consisting of Year, Month, and Day.

A percentage calculation of missing data is performed to determine if there are unfeasible variables. Found missing value which reached 39.6% in the PM2.5 variable, which exceeded the 30% tolerance that can be accepted in the practice of big data science [11]. So, the PM2.5 variable was removed to ensure the data remained appropriate and good to use. Then, replacing unrepresentative characters such as '---', '-', 'N/A' with NaN value. Other empty data is then imputed using the median. The decision to fill in blank data with median exists because it is different from forward fill or backward fill which has the potential to damage the temporal structure of the data, the median is robust and does not depend on the order of observation.

C. Temporal Lag Feature Engineering

The lag variable is used to provide relevant historical information so that the temporal relationship between the PM10 value in the previous period and the current period can be captured. Air quality forecasting because the concentration

of pollutants at one time tends to be affected by previous time conditions, therefore this approach is common to use [12].

Three lag features are formed from PM10 columns for each monitoring station independently. To ensure accurate temporal continuity. To accommodate the autoregressive nature of ISPU time series data, historical pollutant information is represented through engineering temporal lag features.

- t-1: PM10 concentration one day before, representing daily emission memory and undispersed pollutant residuality.
- t-2: PM10 concentration two days in advance, capturing persistent residual pollutant effect for two consecutive days.
- t-7: PM10 concentration seven days before, representing weekly memory capturing the rhythmic cycle of urban activity and weekly commuter patterns.

The establishment of lag features is important to maintain data consistency and prevent information contamination between stations that can cause bias in the modeling process. Feature formation is carried out after the data is grouped based on monitoring stations so that each observation only utilizes historical information from the same station. [13]

D. Independent T-Test Test for Structural Change Analysis

Independent T-Test (two-sample t-test) is a parametric statistical test used to evaluate whether there are significant differences between the averages of two independent sample groups. Comparative environmental studies tend to often use these tests to identify changes in conditions before and after an intervention or time event [14]. Test statistics are formulated as:

$$t = \frac{X_1 - X_2}{s_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} \qquad (1)$$

Where X_1 and X_2 is the average of each group. n_1 dan n_2 is the sample size of each group. s_p is the combined variance calculated using the formula:

$$s_p = \sqrt{\frac{(n_1-1)s_1^2+(n_2-1)s_2^2}{n_1+n_2-2}} \qquad (2)$$

Zero hypothesis (H_0) states that there was no significant difference in the average PM10 concentration between the two periods, while the alternative hypothesis (H_1) states a significant average difference. Test decisions are made based on value p-value at the level of significance $\alpha = 0{,}05$ where H_0 rejected if p-value < 0,05. In this research, Independent T-Test used to test the difference in mean PM10 concentrations between the pre-pandemic period of COVID-19, and the period during/after the pandemic. [14]

The sharing of data into the two temporal regimes allows statistical evaluation of the impact of the pandemic on air quality, particularly PM10 concentrations, through comparison of conditions before and after the intervention [15]. Adopt framework natural experiment, where COVID-19 is treated as an exogenous intervention that causes major changes in population mobility transportation activities and urban economic activities [16].

E. Modeling with Random Forest

Random Forest classify information by recognizing patterns contained within Random Forest used as a model ensemble based on sampling build decision tree simultaneously, by minimizing predictive variations and resisting data interference [17]. Every 'tree' is built using subset, only some features are selected randomly at each branching point (nodes), so that it can produce many diverse trees and reduce relationships with each other, in order to increase the generalization ability of the model.

The final prediction for an observation is obtained through the average of the predictive results of the entire tree, by applying the following:

$$\hat{y} = \frac{1}{B}\sum_{b=1}^{B} f_b\,(x) \qquad (3)$$

where $f_b(x)$ represents the prediction of the (b)-th tree (B) denotes the total number of trees. The aggregation process through averaging enables Random Forest to reduce prediction variance and improve generalization capability, making it more resistant to overfitting than a single decision tree [18]. In this study, the model was configured with n_estimators = 100 and random_state = 42 to maintain experimental reproducibility. It should be noted that Random Forest does not inherently model temporal dependencies on time series data, so historical information needs to be represented through explicitly engineered lag features [19].

F. Interpretation of Models with SHAP Algorithm (Explainable AI)

The use of the SHAP approach provides consistent attribution of features and has a strong mathematical foundation. SHAP or also known as Shapley Additive exPlanations is one of the frameworks Explainable Artificial Interlligence (XAI) which adapts the concept of shapley values from cooperative game theory to explain the contribution of each feature to prediction laerning machine models. In the context of individual predictions, SHAP values represent the mean marginal contribution of a feature to the predicted deviation of the expected value of the model as a whole. [20]

SHAP value for features i on observation x defined as:

$$\phi_k = \sum_{S \subseteq F \setminus \{k\}} \frac{|S|!(|F|-|S|-1)!}{|F|!} [f(S \cup \{k\}) - f(S)] \quad (4)$$

With F is the set of all features, S as a subset of featureless features k, $f(S)$ as a prediction the model uses a subset of features S, $f(S \cup \{k\}) - f(S)$ for marginal contribution of features k on subsets S, Factor $\frac{|S|!(|F|-|S|-1)!}{|F|!}$ is the weight to ensure a fair distribution based on the size of the subset. [21]

In use Random Forest in this research, computational TreeExplainer used. TreeExplainer utilizing the internal structure of the tree to calculate Shapley values precisely with much lower computational complexity than approaches with model-agnostic, thus allowing efficient interpretation even on large tree ensembles such as Random Forest. [21]

G. Evaluation of Models with K-Fold Cross-Validation

In this approach, dataset divided into five subset, with relatively similar measures, in order to obtain stable model performance estimates and represent generalizability to data that has not been observed before, this study applies 5-fold Cross-Validation method. One fold (folds) are used as validation data, while four fold others are used as training data throughout fold get the opportunity to become validation data. The final performance value is then calculated as the average of the evaluation results of all iterations, resulting in more estimates robust compared to the use of one-time training data and test data sharing. [10]

Performance the model was evaluated using four main metrics, namely Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), Mean Absolute Error (MAE), Coefficient of Determination (R^2). The values of the four metric gauges are calculated using the following equation:

$$MSE = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2} \quad (5)$$

$$MAE = \frac{1}{n} \sum_{i=1}^{n} |y_i - \hat{y}_i| \quad (6)$$

$$MAPE = \frac{100\%}{n} \sum_{i=1}^{n} \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (7)$$

Where y_i is the actual value, $\hat{y}_i$ is the predicted value, n is the amount of data. Note, MAPE cannot be used if there is an actual value that has zero value, because there will be a division by zero.[22]

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST} \quad (8)$$

Where SSR (Sum of Squares Regression) is the sum of squares due to regression, measuring the variation described by the model. SSE (Sum of Squares Error) is the sum of the residual squares, measuring the unexplained variation. SHH (Total Sum of Squares) is the total variation of the observed data. [22]

H. Long Term Projections

Model Random Forest the trained, and validated ones were used to project long-term PM10 concentrations through the end of 2027, in order to expand the benefits of the model. By applying methodology iterative auto-regression (or multi-step ahead rolling forecast). The predicted PM10 value results at time t will be entered back into the system to update the temporal lag feature (t-1, t-2, and t-7) and in the next iteration. This process will be repeated sequentially in a loops until all projection target time windows (January 2026 to January 2028) are met. Testing this scenario aims to understand and map potential future pollution fluctuations based on the historical characteristics that the model has studied. [23]

III. Results and Discussion

A. Analysis of Structural Changes in Jakarta Air Quality

The analysis was carried out by comparing conditions during and after COVID-19 (2020-2025) and before the COVID 19 pandemic (2016-2019) using an average difference test (Independent Sample T-Test) to carry out structural change analysis (structural break) in Jakarta air quality, using PM10 concentration data in the 2016-2025 period.

TABLE 2 T-Test Parameters and Values

Parameters	Value
Mean PM10 Before the Pandemic	53.03
Post-Pandemic Mean PM10	51.86
t-statistics	3.6502
p-value	0.00026368355296359225

Table 2 shows the average difference test results (Independent Sample T-Test). Based on the provisions, if the significance level is 5% ($\alpha = 0.05$), a decision is obtained that:

- If p-value < 0.05 then there are significant structural changes.
- If p-value ≥ 0.05 then there are no significant structural changes.

In this research it was obtained p-value of 0.00026368355296359225, this value is smaller than the significance level of 0.05. So it can be concluded that there will be significant structural changes in Jakarta's air quality after 2020.

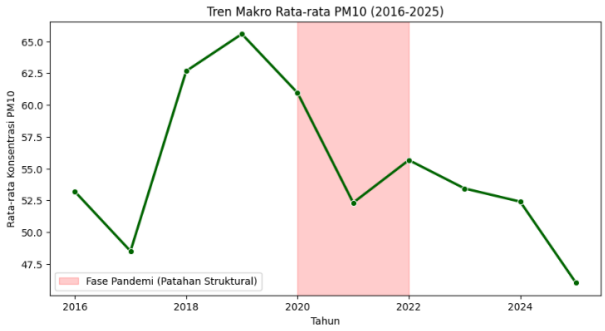


Figure 2 Visualization of Structural Change

Figure 2 shows structural change conditions, with the aim of explaining that Jakarta's air quality data is clearly experiencing structural changes, so the prediction model needs to adapt to changes in data distribution to produce changes in community activity and a decrease in transportation activity, and the division of industrial activities during the pandemic is thought to be the main factor causing changes in PM10 concentration patterns. Based on these calculations, it shows that COVID-19 not only impacts social and economic activities but also Jakarta's air quality.

B. Random Forest Forecasting Performance Evaluation

After testing of structural changes due to the pandemic was verified and identified, the use of random Forest Model developed to predict PM10 concentrations based on lag-based temporal features formed through the .shift() function on the pandas, namely PM10(t-1), PM10(t-2), PM10(t-7). This combination of features was chosen to capture the effects of short-term dependence (previous day) as well as the effects of weekly patterns (weekly seasonality).

1) Accuracy Metric Evaluation

The predictive capacity of the model is measured using a metric comparison of errors against test data sets (test set).

TABLE 3 Test Dataset Error Metrics and Values

Metric	Value
Mean Absolute Error (MAE)	9.34
Root Mean Squared Error (RMSE) Models	12.19
Rmse Baseline	5.46pm
Coefficient of Determination (R^2)	0.41
Mean Absolute Percentage Error (MAPE)	72.77%

Table 3 shows the model produces an RMSE value of **12.19** and the R^2 value is **0.41.** This value shows that Random Forest able to explain most of the variation in PM10 concentration in test data. Relatively high MAPE values are influenced by the presence of low value PM10 observations, resulting in errors a small absolute can produce a large percentage of errors. This fact shows that Random Forest effectively used to capture non-linear characteristics of air quality data that tend to be influenced by many factors and are complex. The RMSE model is lower compared to the RMSE baseline, meaning that the model manages to study inter-variable relationship patterns better when compared with a simple prediction approach. Visualization **Figure 3** will show a comparison of actual and predicted PM10 Random Forest throughout 2025.

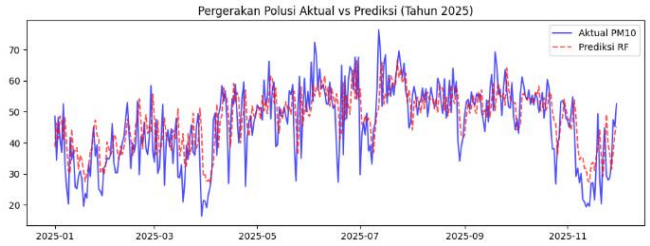


Figure 3 Comparison of Actual Values and PM10 Predictions

2) Residual Analysis (Distribution and Autocorrelation Error)

A model is said to be statistically healthy"if error the resulting nature white noise, that is, the remaining errors are random, normally distributed, and do not store any patterns of information that are missed.

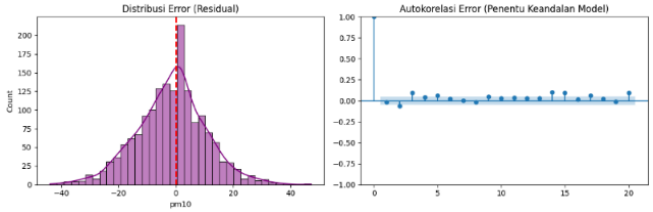


Figure 4 Distribution Visualization and Error Autocorrelation

Figure 4 showing the symmetrical bell formed from the left image presents a distribution curve error which is centered on the number zero. This indicates the absence of systematic bias, the model is not consistent underpredict or overpredict, but rather do the deviation in a balanced manner. Autocorrelation Function (ACF) which is visualized through the right image, explains that the correlation betweenlag in residuals it falls close to the zero line and is inside the blue significance band. This explains that the model of most patterns that the model can learn has been successfully extracted, while the remaining residuals tend to be random.

3) Validity Stability Model (Time-Series Cross-Validation)

To ensure that the model's performance does not experience overfitting and has high generalization ability in different time windows, stability tests are carried out using methods TimeSeriesSplit (n_splits=5). Serving to maintain the chronological order of the data when cross validation is performed, this step is one of the crucial parts.

TABLE 4 Cross-Validation Time-Series Test

Time Fold (Fold)	Rmse score	Score R^2

Fold 1	13.94463925	0.36091023
Fold 2	12.32110206	0.31505399
Fold 3	12.05927598	0.33615718
Fold 4	11.60947278	0.42034806
Fold 5	12.00138879	0.41116825
Stable Average	**12.39**	**0.37**

Table 4 showing the results of the model stability test shows an average RMSE value of 12.39, and an average of R^2 of 0.37. The consistency of metric values in each of these time folds confirms that the level of error the model remains stable and reliable even though it is tested at varying historical time segments.

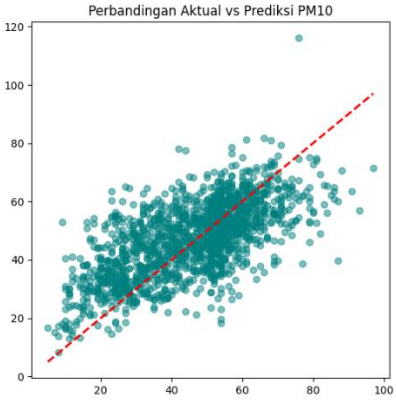


Figure 5 Cross-Validation Time-Series Test Visualization

Figure 5 shows the nature of the model Random Forest tends to produce more conservative predictions (close to the average value of the training data distribution), so deviations are still found in the form of under-prediction at peaks of extreme values (very high pollution spikes). However, in the figure, it can be seen that the predicted outcome chart is able to follow daily fluctuations and actual data trend patterns consistently.

C. Explainable Artificial Intelligence (XAI) Analysis

To increase transparency random Forest Model, an approach is used Explainable Artificial Intelligence (XAI) with the SHAP method (SHapley Additive exPlanations) who is able to explain the contribution of each feature to the predicted results.

1) Global Feature Importance

TABLE 5 Temporal Feature Score Test

Ranking	Temporal Feature Score	Mean SHAP
1	PM10 (t-1)	7.768793
2	PM10 (t-7)	2.352499
3	PM10 (t-2)	1.869012

Table 5 showed that the t-1 feature proved to make the largest contribution globally to model predictions. This analysis shows the existence of traits temporal peristence which is very strong in Jakarta's air atmosphere, with the level of pollutant accumulation in the previous day holding dominant control in projecting pollution concentrations in the next day. **Figure 6** below visualizes the magnitude of the feature contribution to the model:

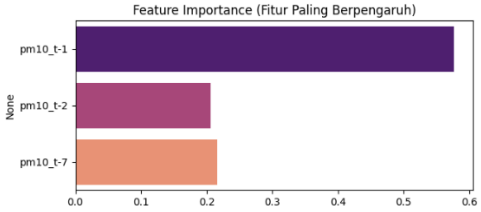


Figure 6 Global Feature Importance Visualization

2) SHAP Summary Plot

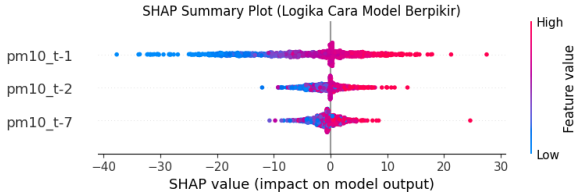


Figure 7 SHAP Summary Plot Visualization

Figure 7 showing visualization results shows that high PM(t-1) tends to produce a positive SHAP value. This means that the higher the PM10 concentration on the previous day, the greater the contribution in improving PM10 predictions on the following day. On the other hand, if PM10 (t-1) is low, it tends to make a negative contribution to lowering the model's prediction results. This pattern explains that the current relationship of PM10 to PM10 one day before is strong and stable.

3) Non-Linear Dependency Analysis

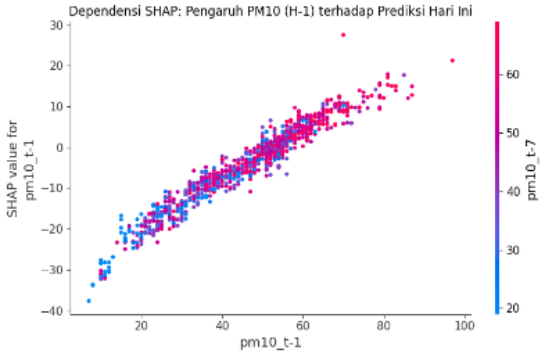


Figure 8 Visualization of Non-Linear Dependencies

Figure 8 shows that the relationship between PM10(t-1) and the predicted value is not completely linear. This phenomenon explains that Random Forest successfully

capturing non-linear interactions that are difficult to model using conventional linear regression approaches. In the low to medium value range, an increase in PM10 (t−1) causes a gradual increase in SHAP contribution. However, at higher values, the rate of increase in contributions is not always constant and shows a more complex pattern. The ability to capture non-linear relationships is one of the main advantages Random Forest in the context of air quality forecasting. Patterns of relationships found through SHAP Dependence Plot reinforces the fact that models are able to represent air quality dynamics more realistically.

D. Model Resilience Test (What-If Analysis)

Simulation test stress done through What-If Analsis to validate the heuristic logic of the model. This manual predictive approach forces the model to respond to extreme man-made scenarios that are not necessarily present in historical data, to ensure that the model not only memorizes old data patterns, but understands temporal between PM10 lag features.

TABLE 6 Hypothesis Scenarios and Input Manipulation

Hypothesis Scenario	Input Manipulation (pm10_t-1, t-2, t-7)	Mean SHAP
Baseline Scenario	L1=50, L2=48, L7=50	46.627778
Extreme Disaster Scenario	L1=150, L2=48, L7=50	75.080000
Clean Wind Sweep Scenario	L1=20, L2=48, L7=50	40.120000

Table 6 shows when the model is given the scenario "Extreme Disaster" with past pollution levels (t-1) forced to jump to 150$\mu g/m3$, the results show a significant increase. However, when the scenario is "Clean Wind Sweep" with 20$\mu g/m3$ models project a drastic reduction in pollution. Scenario values were selected based on minimum and maximum observation ranges PM10 found in historical datasets. Also carried out was an analysis of the impact of mass policies which previously showed that the 20% pollution reduction intervention the previous day had a systemic impact on improving air quality today. Overall, the model has demonstrated a consistent predictive response to changes in input scenarios, making it feasible to use it as a basis for strategic decision-making related to air quality.

E. Long Term Projections of Air Quality (Forecasting 2026-2027)

Utilizing methodology iterative auto-regression, namely the predicted results are processed back into lag features in the next prediction step, the model projects Jakarta air quality until the end of 2027.

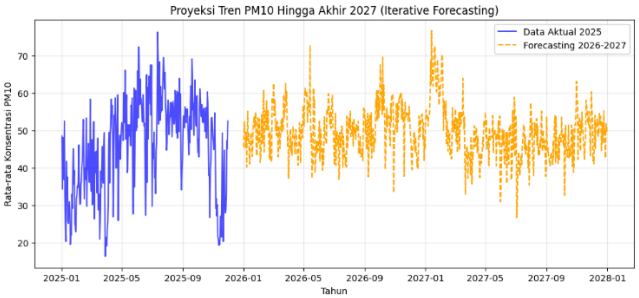


Figure 9 Long Term Projections

The image shows projections of several periods with relatively high PM10 concentrations based on historical patterns studied. Despite daily fluctuations, the general pattern that has formed shows that Jakarta's air quality conditions are still strongly influenced by its historical characteristics. This projection is a simulation based on historical patterns and does not consider external changes such as environmental policy, meteorological conditions, or changes in economic activity. These findings show that models are not only able to provide short-term predictions, but can also be used as a decision support tool (decision support tool) in planning air pollution control policies.

IV. Conclusions and Suggestions

Based on the results of analysis of Jakarta air quality data for the 2016-2025 period, this research has succeeded in achieving the set goals, by building a model that is not only able to predict PM10-based Random Forest but it can also be explained through Explainable Artificial Intelligence (XAI). Utilization Independent Sample T-Test indicating a significant structural change in air quality in Jakarta after the 2020 datum, this is proven by the value p-value obtained amounting to 0.00026368355296359225 (< 0.05). The results of this analysis strengthen the assumption that COVID-19 has an effect on changes in the distribution pattern of PM10 concentrations due to changes in community mobilization and urban economic activity.

The resulting RMSE was 12.19, with MAE of 9.34, and determination coefficient (R^2) of 0.41 indicate that the model Random Forest which was built using the temporal lag features PM10(t-1), PM10(t-2), and PM10(t-7) is able to produce quite good performance. In various time segments with an average RMSE of 12.39 and an average R^2 of 0.3 validation results obtained using Time-Series Cross-Validation also shows a consistent level of stability. Residual analysis also showed that prediction errors were random and did not leave significant autocorrelation patterns, indicating that the model had succeeded in capturing most of the information contained in historical data.

SHAP dependency analysis explains the existence of a non-linear relationship between historical PM10 concentrations and model predictions. The application of the SHAP method as the XAI approach succeeded in explaining the contribution of each feature to the predicted results

obtained Random Forest so it is able to increase model transparency by explaining the contribution of each feature to predictions. This indicates Random Forest's ability to capture the complexity of urban air quality dynamics. Implementation what-if analysis demonstrates the ability of the model to provide logical responses in various hypothesis scenarios so that it has the potential to be used as a support tool for decision makers in air quality management.

For future research, it is highly recommended to carry out comparisons with other time series methods, such as Long Short-Term Memory (LSTM), Temporal Fusion Transformer (TFT), as well as other models. Also, the model was developed to include other external variables such as rainfall, wind speed, air temperature, and others.

THANK YOU

The author would like to thank the lecturer who taught the Big Data course, Mr. Tony S.Kom., M.Kom., Ph. D. who has provided direction, input and guidance during the implementation of this research. All relatives, friends, senior brothers, helped provide direction and guidance and support in this research. As well as unconditional data providers who have provided data and become the basis for the theoretical basis in this research so that this research can be carried out well. Hopefully this research can be useful for developers and other readers.

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